**Stat 6390.001: High Dimensional Probability and Statistics, Fall 2026** **Topic 1**

# High Dimensional Probability and Statistics

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# Chapter 1

# Basic tail and concentration bounds

## 1.1 Introduction

A central challenge in high-dimensional statistics is the *curse of dimensionality*. As the dimension $d$ grows, the sample space expands rapidly, so a fixed number of observations becomes increasingly sparse relative to the size of the space. As a result, methods and intuition that work well in low dimensions may become much less informative in high-dimensional settings.

High-dimensional problems also exhibit geometric and probabilistic behavior that can differ substantially from low-dimensional intuition. For example, if

$$X \sim N(0, I_d),$$

then the density is maximized at the origin, but a typical sample is not concentrated near the origin. Instead,

$$\|X\|_2 \approx \sqrt{d},$$

so most samples lie near a thin shell of radius approximately $\sqrt{d}$.

These phenomena motivate the study of probabilistic tools that remain useful when the dimension is large. In particular, this lecture develops tail and concentration bounds that quantify how likely a random variable is to deviate from its mean, with an emphasis on non-asymptotic guarantees.

### 1.1.1 One-dimensional and multivariate CLT

Let $X_1, \dots, X_n$ be i.i.d. random variables with mean $\mu$ and finite variance $\sigma^2$. The one-dimensional CLT states that

$$\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n} X_i - \mu\right) \xrightarrow{d} N(0, \sigma^2) \qquad \text{as } n \to \infty. \tag{1.1}$$

The multivariate version has the same form. Suppose now that $X_1, \dots, X_n$ are i.i.d. random vectors in $\mathbb{R}^d$ with

$$\mathbb{E}[X_i] = \mu \in \mathbb{R}^d, \qquad \mathrm{Cov}(X_i) = \Sigma.$$

Then

$$\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n} X_i - \mu\right) \xrightarrow{d} N(0, \Sigma) \qquad \text{as } n \to \infty. \tag{1.2}$$

For a $d$-dimensional Gaussian vector $X \sim N(\mu, \Sigma)$ with positive definite covariance matrix $\Sigma$, the density is

$$f(x) = \frac{1}{(2\pi)^{d/2}|\Sigma|^{1/2}} \exp\left\{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right\}. \tag{1.3}$$

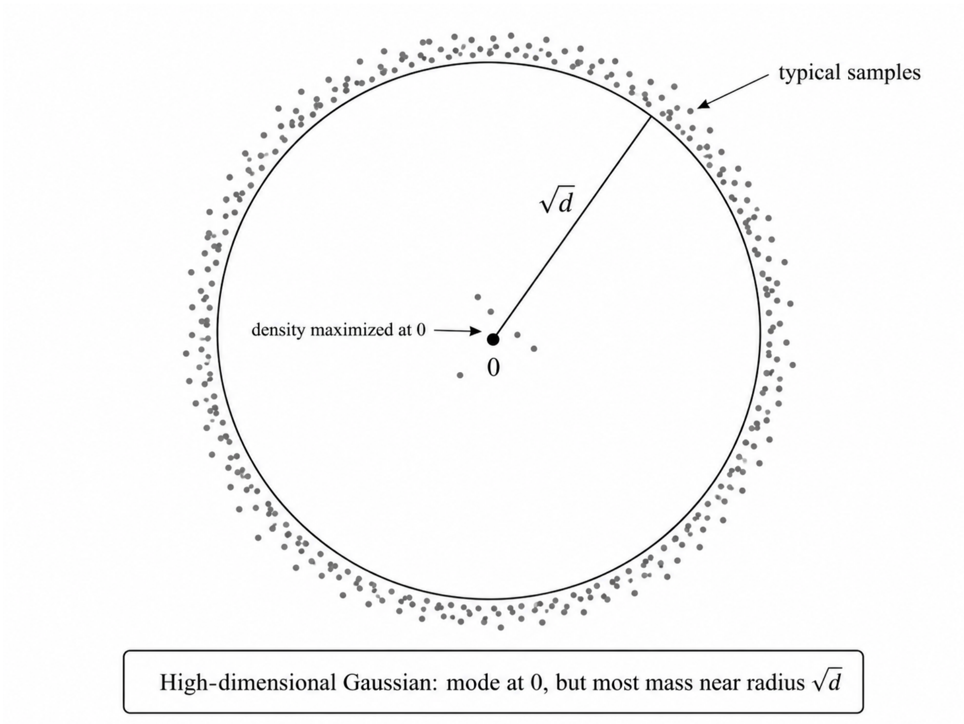


Figure 1.1: A schematic illustration of a high-dimensional standard Gaussian distribution.

**Example 1.1.1** (A First High-Dimensional Phenomenon)**.** Consider the standard Gaussian case $X \sim N(0, I_d)$. The density is maximized at the origin, so the mode is $x = 0$. However, when $d$ is large, a typical draw is not close to the origin. Indeed,

$$\mathbb{E}\|X\|_2^2 = \mathbb{E}\left[\sum_{j=1}^{d} X_j^2\right] = \sum_{j=1}^{d} \mathbb{E}[X_j^2] = d. \tag{1.4}$$

Thus the typical Euclidean radius is of order $\sqrt{d}$; more precisely, $\|X\|_2^2$ is a chi-square random variable with $d$ degrees of freedom and is concentrated around $d$. Hence most of the Gaussian probability mass lies in a thin shell with radius approximately $\sqrt{d}$, even though the density itself is largest at the origin, shown in Figure 1. This illustrates a basic feature of high-dimensional geometry: the location of the mode need not describe where a typical sample lies.

## 1.2 Classical Tail Bounds

The purpose of a tail bound is to control probabilities such as $\mathbb{P}(X \geq t)$ or $\mathbb{P}(|X - \mathbb{E}X| \geq t)$. The amount of information available about the distribution determines the quality of the bound. Markov's inequality uses only a first moment; Chebyshev's inequality uses a second moment; and the Chernoff method exploits the moment generating function (MGF), thereby incorporating information from all moments when the MGF exists [1].

### 1.2.1 Markov and Chebyshev inequalities

**Proposition 1.2.1** (Markov's inequality)**.** *If random variable $X$ is nonnegative and $\mathbb{E}[X] < \infty$, then for* *every* $t > 0$*,*

$$\mathbb{P}(X \geq t) \leq \frac{\mathbb{E}[X]}{t}. \tag{1.5}$$

**Proposition 1.2.2** (Chebyshev's inequality)**.** *If random variable* $X$ *has finite variance and* $\mu = \mathbb{E}[X]$*, then for every* $t > 0$*,*

$$\mathbb{P}\big(|X - \mu| \geq t\big) \leq \frac{\mathrm{Var}(X)}{t^2}. \tag{1.6}$$

Chebyshev's inequality is an immediate consequence of Markov's inequality applied to the nonnegative random variable $(X - \mu)^2$. More generally, if the $k$th centered absolute moment exists, then

$$\mathbb{P}\big(|X - \mu| \geq t\big) = \mathbb{P}\big(|X - \mu|^k \geq t^k\big) \leq \frac{\mathbb{E}|X - \mu|^k}{t^k}. \tag{1.7}$$

### 1.2.2 From Markov to the Chernoff bound

The exponential function is particularly effective because it converts sums into products and leads naturally to the moment generating function.

**Lemma 1.2.3** (Chernoff bound)**.** *Let* $\mu = \mathbb{E}[X]$*, and suppose*

$$\phi(\lambda) = \mathbb{E}\big[e^{\lambda(X-\mu)}\big]$$

*exists for* $\lambda \in [0, b]$*. Then for every* $t > 0$*,*

$$\log \mathbb{P}(X - \mu \geq t) \leq \inf_{\lambda \in [0,b]} \left\{ \log \phi(\lambda) - \lambda t \right\}. \tag{1.8}$$

*Proof.* Fox $\lambda > 0$. Since the exponential function is increasing,

$$\mathbb{P}(X - \mu \geq t) = \mathbb{P}\big(\lambda(X - \mu) \geq \lambda t\big) \tag{1.9}$$

$$= \mathbb{P}\big(e^{\lambda(X-\mu)} \geq e^{\lambda t}\big). \tag{1.10}$$

Applying Markov's inequality to the nonnegative random variable $e^{\lambda(X-\mu)}$ gives

$$\mathbb{P}(X - \mu \geq t) \leq \frac{\mathbb{E}[e^{\lambda(X-\mu)}]}{e^{\lambda t}} = \phi(\lambda) e^{-\lambda t}. \tag{1.11}$$

Taking logarithms yields

$$\log \mathbb{P}(X - \mu \geq t) \leq \log \phi(\lambda) - \lambda t.$$

Because the inequality holds for every admissible $\lambda$, minimizing the right-hand side over $\lambda \in [0, b]$ gives the result. □

**Remark 1.2.4.** The optimization over $\lambda$ is essential. Different values of $\lambda$ produce different valid bounds, and the infimum selects the sharpest one obtained from this argument. In applications, the main work is therefore to control the MGF and then solve a one-dimensional optimization problem.

## 1.3 Sub-Gaussian Random Variables

### 1.3.1 Gaussian Tail Bound

As a first example, consider a Gaussian random variable

$$X \sim N(\mu, \sigma^2).$$

Its moment generating function is

$$\mathbb{E}[e^{\lambda X}] = \exp\left(\mu\lambda + \frac{\sigma^2\lambda^2}{2}\right), \qquad \lambda \in \mathbb{R}. \tag{1.12}$$

Applying the preceding lemma, the upper-tail bound is determined by

$$\inf_{\lambda>0}\left\{\log \mathbb{E}[e^{\lambda(X-\mu)}] - \lambda t\right\} = \inf_{\lambda>0}\left\{\frac{\sigma^2\lambda^2}{2} - \lambda t\right\}. \tag{1.13}$$

Here we used the fact that

$$X - \mu \sim N(0, \sigma^2).$$

The quadratic expression is minimized at

$$\lambda^* = \frac{t}{\sigma^2},$$

which gives

$$\inf_{\lambda>0}\left\{\frac{\sigma^2\lambda^2}{2} - \lambda t\right\} = -\frac{t^2}{2\sigma^2}. \tag{1.14}$$

Therefore, for every $t > 0$,

$$\mathbb{P}(X - \mu \geq t) \leq \exp\left(-\frac{t^2}{2\sigma^2}\right). \tag{1.15}$$

(Normal distribution is light tail)

### 1.3.2 Definition and basic tail bounds

**Definition 1.3.1** (Sub-Gaussian random variable)**.** A random variable $X$ with mean $\mu = \mathbb{E}[X]$ is *sub-Gaussian with parameter* $\sigma > 0$ if

$$\mathbb{E}\left[e^{\lambda(X-\mu)}\right] \leq \exp\left(\frac{\sigma^2\lambda^2}{2}\right) \qquad \text{for all } \lambda \in \mathbb{R}. \tag{1.16}$$

**Remark 1.3.2.** If $X$ is $\sigma$-sub-Gaussian for some $\sigma > 0$, then $X$ is also $\sigma'$-sub-Gaussian for some $\sigma' \geq \sigma$.

The definition says that the centered MGF of $X$ grows no faster than the centered MGF of a Gaussian $N(0, \sigma^2)$. Therefore, the same Chernoff calculation gives the Gaussian-type upper-tail bound

$$\mathbb{P}(X - \mu \geq t) \leq \exp\left(-\frac{t^2}{2\sigma^2}\right), \qquad t > 0. \tag{1.17}$$

**Lemma 1.3.3** (Symmetry of sub-Gaussianity)**.** *A random variable $X$ is sub-Gaussian with parameter $\sigma$ if and only if $-X$ is sub-Gaussian with the same parameter.*

*Proof.* Let $\mu = \mathbb{E}[X]$. Recall that $X$ is sub-Gaussian with parameter $\sigma$ if

$$\mathbb{E}\left[e^{\lambda(X-\mu)}\right] \leq \exp\left(\frac{\sigma^2\lambda^2}{2}\right), \qquad \lambda \in \mathbb{R}. \tag{1.18}$$

Suppose first that $X$ is sub-Gaussian. Since $\mathbb{E}[-X] = -\mu$,

$$\begin{aligned}
\mathbb{E}\left[e^{\lambda((-X)-\mathbb{E}[-X])}\right] &= \mathbb{E}\left[e^{-\lambda(X-\mu)}\right] && (1.19)\\
&\leq \exp\left(\frac{\sigma^2(-\lambda)^2}{2}\right) && (1.20)\\
&= \exp\left(\frac{\sigma^2\lambda^2}{2}\right). && (1.21)
\end{aligned}$$

Hence $-X$ is sub-Gaussian with parameter $\sigma$.

Conversely, suppose that $-X$ is sub-Gaussian. Then

$$\mathbb{E}\left[e^{\lambda(X-\mu)}\right] = \mathbb{E}\left[e^{-\lambda((-X)-\mathbb{E}[-X])}\right] \tag{1.22}$$
$$\leq \exp\left(\frac{\sigma^2(-\lambda)^2}{2}\right) \tag{1.23}$$
$$= \exp\left(\frac{\sigma^2\lambda^2}{2}\right). \tag{1.24}$$

Therefore $X$ is sub-Gaussian with parameter $\sigma$. □

**Lemma 1.3.4** (Lower Tail)**.** *If $X$ is sub-Gaussian with parameter $\sigma$, then for every $t > 0$,*

$$\mathbb{P}(X - \mathbb{E}[X] \leq -t) \leq \exp\left(-\frac{t^2}{2\sigma^2}\right). \tag{1.25}$$

*Proof.* By the symmetry lemma, $-X$ is also sub-Gaussian with parameter $\sigma$. Moreover,

$$\mathbb{P}(X - \mathbb{E}[X] \leq -t) = \mathbb{P}(-X - \mathbb{E}[-X] \geq t) \tag{1.26}$$
$$\leq \exp\left(-\frac{t^2}{2\sigma^2}\right), \tag{1.27}$$

where the last step follows from the upper-tail bound applied to $-X$. □

Now if we combine upper and lower tails with the union bound yields the two-sided concentration inequality we will get the following

**Lemma 1.3.5.** *If $X$ is sub-Gaussian with parameter $\sigma > 0$,*

$$\mathbb{P}\big(|X - \mathbb{E}X| \geq t\big) \leq 2\exp\left(-\frac{t^2}{2\sigma^2}\right), \qquad t > 0. \tag{1.28}$$

Note: This much faster decay is one reason sub-Gaussian assumptions are central in non-asymptotic high-dimensional statistics [1].

**Example 1.3.6** (Rademacher random variable)**.** A Rademacher random variable $\operatorname{Var} epsilon$ takes values in $\{-1, +1\}$ equiprobably:

In particular, $\mathbb{E}[\operatorname{Var} epsilon] = 0$. Its MGF is

$$\mathbb{E}[e^{\lambda \operatorname{Var} epsilon}] = \frac{1}{2}e^{\lambda} + \frac{1}{2}e^{-\lambda} \tag{1.29}$$
$$= \frac{1}{2}\left(\sum_{k=0}^{\infty}\frac{\lambda^k}{k!} + \sum_{k=0}^{\infty}\frac{(-\lambda)^k}{k!}\right) \tag{1.30}$$
$$= \sum_{m=0}^{\infty}\frac{\lambda^{2m}}{(2m)!}, \tag{1.31}$$

where the odd-order terms cancel.

Since $(2m)! \geq 2^m m!$,

$$\mathbb{E}[e^{\lambda \operatorname{Var} epsilon}] \leq \sum_{m=0}^{\infty}\frac{\lambda^{2m}}{2^m m!} \tag{1.32}$$
$$= e^{\lambda^2/2}. \tag{1.33}$$

Since $\mathbb{E}[\operatorname{Var} epsilon] = 0$, it follows that

$$\mathbb{E}[e^{\lambda(\operatorname{Var} epsilon - \mathbb{E}[\operatorname{Var} epsilon])}] \leq e^{\lambda^2/2}.$$

Therefore, $\operatorname{Var} epsilon$ is sub-Gaussian with parameter $\sigma = 1$.

This example shows that sub-Gaussianity is not the same as Gaussianity. A Rademacher variable is discrete and bounded, yet its MGF is dominated by that of a standard Gaussian. The sub-Gaussian class is therefore broad enough to include many non-Gaussian random variables while retaining Gaussian-like tail behavior.

Note:"equiprobably" meaning

$$\mathbb{P}(\operatorname{Var} epsilon = 1) = \mathbb{P}(\operatorname{Var} epsilon = -1) = \frac{1}{2}. \tag{1.34}$$

Therefore,

$$\mathbb{E}[\operatorname{Var} epsilon] = \frac{1}{2}(1) + \frac{1}{2}(-1) = 0. \tag{1.35}$$

### 1.3.3 Example of Sub-Gaussian Variables

**Example 1.3.7** (Bounded random variables)**.** Let $X$ be zero-mean, and supported on some interval $[a, b]$. We will show that $X$ is sub-Gaussian with parameter at most $\sigma = b - a$.

Letting $X'$ be an independent copy of $X$, so that $X \stackrel{d}{=} X'$ and $X \perp\!\!\!\perp X'$. For any $\lambda \in \mathbb{R}$, we have

$$\mathbb{E}_X\left[e^{\lambda X}\right] = \mathbb{E}_X\left[e^{\lambda(X - \mathbb{E}_{X'}[X'])}\right] \stackrel{\text{(i)}}{\leq} \mathbb{E}_X\left[\mathbb{E}_{X'} e^{\lambda(X - X')}\right] = \mathbb{E}_{X,X'}\left[e^{\lambda(X-X')}\right]$$

The first equality is because $\mathbb{E}[X'] = 0$, and (i) follows from the the convexity of the exponential, and Jensen's inequality. Letting $\varepsilon$ be an independent Rademacher variable, note that since $(X, X')$ are i.i.d, $X - X' \stackrel{d}{=} X' - X$, so $(X - X')$ is symmetric about 0. Thus, the distribution of $(X - X')$ is the same as that of $\varepsilon(X - X')$, so that we have

$$\mathbb{E}_{X,X'}\left[e^{\lambda(X-X')}\right] = \mathbb{E}_{X,X'}\left[\mathbb{E}_\varepsilon\left[e^{\lambda\varepsilon(X-X')}\right]\right] \stackrel{\text{(ii)}}{\leq} \mathbb{E}_{X,X'}\left[e^{\frac{\lambda^2}{2}(X-X')^2}\right] \leq e^{\frac{\lambda^2}{2}(b-a)^2}$$

The first equality follows from Tonelli's theorem and (ii) follows from the sub-Gaussianity of Radamacher variable and last inequality is because $|X - X'| \leq b - a$. Putting everything together, we have shown that $X$ is sub-Gaussian with parameter at most $\sigma = b - a$.

**Remark 1.3.8.** This result is useful, but it can be sharpened. It can be shown that $X$ is sub-Gaussian with parameter at most $\sigma = \frac{b-a}{2}$ (Exercise 2.4). The technique used in this example is a simple example of a *symmetrization argument*, in which we first introduce an independent copy $X'$, and then symmetrize the problem with a Rademacher variable. Such symmetrization arguments are useful in a variety of contexts.

### 1.3.4 Hoeffding's Inequality

Another property of sub-Gaussianity is preserved by linear operations.

**Lemma 1.3.9.** *Let $X_1$ and $X_2$ be independent sub-Gaussian variables with parameters $\sigma_1$ and $\sigma_2$, respectively, then $X_1 + X_2$ is sub-Gaussian with parameter $\sqrt{\sigma_1^2 + \sigma_2^2}$.*

*Proof.* By independence we have

$$\mathbb{E}\left[e^{\lambda(X_1+X_2-\mathbb{E}X_1-\mathbb{E}X_2)}\right] = \mathbb{E}\left[e^{\lambda(X_1-\mathbb{E}X_1)}\right]\mathbb{E}\left[e^{\lambda(X_2-\mathbb{E}X_2)}\right] \le e^{\frac{\lambda^2}{2}\sigma_1^2}e^{\frac{\lambda^2}{2}\sigma_2^2} = e^{\frac{\lambda^2}{2}(\sigma_1^2+\sigma_2^2)}$$

□

As a consequence of this fact, we obtain an important result known as the *Hoeffding's inequality*:

**Proposition 1.3.10** (Hoeffding's inequality)**.** *Suppose that the variables $X_i$, $i = 1, \dots, n$, are independent, and $X_i$ has mean $\mu_i$ and sub-Gaussian parameter $\sigma_i$. Then for all $t \ge 0$, we have*

$$\mathbb{P}\left[\sum_{i=1}^n (X_i - \mu_i) \ge t\right] \le \exp\left\{-\frac{t^2}{2\sum_{i=1}^n \sigma_i^2}\right\}$$

*and*

$$\mathbb{P}\left[\left|\sum_{i=1}^n X_i - \sum_{i=1}^n \mathbb{E}X_i\right| \ge t\right] \le 2\exp\left\{-\frac{t^2}{2\sum_{i=1}^n \sigma_i^2}\right\}$$

*Proof.* By previous lemma, $\sum_{i=1}^n X_i$ is $\sqrt{\sigma_1^2 + \cdots + \sigma_n^2}$ sub-Gaussian. So, by basic sub-Gaussian tail bound we have the results. □

**Corollary 1.3.11.** *If $X_i$, $i = 1, \dots, n$, are i.i.d, with mean $\mu$ and sub-Gaussian parameter $\sigma$. Then for all $t \ge 0$, we have*

$$\mathbb{P}\left[\left|\frac{1}{n}\sum_{i=1}^n X_i - \mu\right| \ge t\right] \le 2\exp\left\{-\frac{nt^2}{2\sigma^2}\right\}$$

*and*

$$\mathbb{P}\left[\sqrt{n}\left|\frac{1}{n}\sum_{i=1}^n X_i - \mu\right| \ge t\right] \le 2\exp\left\{-\frac{t^2}{2\sigma^2}\right\}$$

**Remark 1.3.12.** The above can be viewed as a finite sample Central Limit Theorem. CLT states that when $n \to \infty$,

$$\sqrt{n}\left|\frac{1}{n}\sum_{i=1}^n X_i - \mu\right| \to \mathcal{N}(0, \sigma^2).$$

The second statement of the above corollary is a finite sample version of CLT, by stating that the tail of $\sqrt{n}\left|\frac{1}{n}\sum_{i=1}^n X_i - \mu\right|$ is upper bounded by the tail of $\mathcal{N}(0, \sigma^2)$.

*Proof.* By Hoeffding,

$$\mathbb{P}\left[\left|\frac{1}{n}\sum_{i=1}^n X_i - \mu\right| \ge t\right] = \mathbb{P}\left[\left|\sum_{i=1}^n X_i - n\mu\right| \ge nt\right] \le 2\exp\left\{-\frac{(nt)^2}{2n\sigma^2}\right\} = 2\exp\left\{-\frac{nt^2}{2\sigma^2}\right\}$$

Second statement follows analogously. □

One application of Hoeffding's bound is to derive finite sample guarantees for the error of the sample mean.

**Example 1.3.13** (Finite-sample error guarantee)**.** Let $\overline{X}_n := \frac{1}{n}\sum_{i=1}^n X_i$ and fix a confidence level $\delta \in (0, 1)$. By the preceding corollary,

$$\mathbb{P}\left[\left|\overline{X}_n - \mu\right| \ge t\right] \le 2\exp\left\{-\frac{nt^2}{2\sigma^2}\right\}$$

Choose $t = \sqrt{\frac{2\sigma^2}{n}\log\left(\frac{2}{\delta}\right)}$, so that $\delta = 2\exp\left\{-\frac{nt^2}{2\sigma^2}\right\}$. In other words, with probability at least $1 - \delta$,

$$\left|\overline{X}_n - \mu\right| \le \sqrt{\frac{2\sigma^2}{n}\log\left(\frac{2}{\delta}\right)}$$

**Example 1.3.14** (Expected absolute error)**.** We can also control the expected absolute error $\mathbb{E}\left|\overline{X}_n - \mu\right|$ directly from the tail bound. Recall that for any nonnegative random variable $Y$, Tonelli's theorem gives

$$\mathbb{E}[Y] = \int_0^\infty \mathbb{P}(Y \geq t)\, dt$$

Taking $Y = \left|\overline{X}_n - \mu\right|$, the preceding corollary and the trivial bound $\mathbb{P}(Y \geq t) \leq 1$ imply

$$\mathbb{P}\left[\left|\overline{X}_n - \mu\right| \geq t\right] \leq \min\left\{1,\ 2\exp\left\{-\frac{nt^2}{2\sigma^2}\right\}\right\}$$

Let $t_0$ be the point at which the two upper bounds agree. Thus $2e^{-\frac{nt_0^2}{2\sigma^2}} = 1$, which gives $t_0 = \sigma\sqrt{\frac{2\log 2}{n}}$. Hence

$$\begin{aligned}
\mathbb{E}\left|\overline{X}_n - \mu\right| &= \int_0^\infty \mathbb{P}\left[\left|\overline{X}_n - \mu\right| \geq t\right]\, dt \\
&\leq \int_0^{t_0} 1\, dt + \underbrace{\int_{t_0}^\infty 2\exp\left\{-\frac{nt^2}{2\sigma^2}\right\} dt}_{z = \frac{\sqrt{n}}{\sigma}t, \quad dt = \frac{\sigma}{\sqrt{n}}\, dz} \\
&= t_0 + \frac{2\sigma}{\sqrt{n}} \int_{\frac{\sqrt{n}}{\sigma}t_0}^\infty e^{-z^2/2}\, dz \\
&= \frac{\sigma}{\sqrt{n}}\left(\sqrt{2\log 2} + 2\int_{\sqrt{2\log 2}}^\infty e^{-z^2/2}\, dz\right) \\
&\leq 2\frac{\sigma}{\sqrt{n}}.
\end{aligned}$$

We conclude our discussion of sub-Gaussianity with a theorem that provides different characterizations of sub-Gaussian variables.

**Theorem 1.3.15** (Equivalent characterizations of sub-Gaussian variables)**.** *Given any zero-mean random variable $X$, the following properties are equivalent:*

1. *There is a constant $\sigma \geq 0$ such that*

$$\mathbb{E}\left[e^{\lambda X}\right] \leq e^{\frac{\lambda^2\sigma^2}{2}} \qquad \text{for all } \lambda \in \mathbb{R}.$$

2. *There is a constant $c \geq 0$ and Gaussian random variable $Z \sim \mathcal{N}(0, \tau^2)$ such that*

$$\mathbb{P}\left[|X| \geq s\right] \leq c\,\mathbb{P}\left[|Z| \geq s\right] \qquad \text{for all } s \geq 0.$$

3. *There is a constant $\theta \geq 0$ such that*

$$\mathbb{E}\left[X^{2k}\right] \leq \frac{(2k)!}{2^k k!}\theta^{2k} \qquad \text{for all } k = 1, 2, \ldots$$

4. *There is a constant $\sigma \geq 0$ such that*

$$\mathbb{E}\left[e^{\frac{\lambda X^2}{2\sigma^2}}\right] \leq \frac{1}{\sqrt{1-\lambda}} \qquad \text{for all } \lambda \in [0, 1).$$

*Proof.* See Appendix A (Section 2.4) in [1]. ∎

## 1.4 Sub-Exponential Variables

We now turn to the class of sub-exponential variables, which are defined by a slightly milder condition on the moment generating function.

**Definition 1.4.1.** A random variable $X$ with mean $\mu = \mathbb{E}[X]$ is *sub-exponential* if there are non-negative parameters $(\nu, \alpha)$ such that

$$\mathbb{E}\left[e^{\lambda(X-\mu)}\right] \leq e^{\frac{\nu^2\lambda^2}{2}} \qquad \text{for all } |\lambda| < \frac{1}{\alpha}$$

# Bibliography

[1] Martin J. Wainwright, *High-Dimensional Statistics: A Non-Asymptotic Viewpoint*, Cambridge University Press, 2019.